



Multi-modal feature-fusion for CT metal artifact reduction using edge-enhanced generative adversarial networks[☆]

Zhiwei Huang^{a,b,c}, Guo Zhang^{a,b,c}, Jinzhao Lin^{a,c,*}, Yu Pang^c, Huiqian Wang^c, Tong Bai^c, Lisha Zhong^{a,b,*}

^a School of Communication and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing 400065, China

^b School of Medical Information and Engineering, Southwest Medical University, Luzhou, 646000, China

^c Chongqing Key Laboratory of Photo-electronic Information Sensing and Transmitting Technology, Chongqing University of Posts and Telecommunications, Chongqing, 400065, China

ARTICLE INFO

Article history:

Received 10 June 2021

Revised 6 February 2022

Accepted 11 February 2022

Keywords:

Feature fusion

Generative adversarial network

Metal artifact reduction

Second artifact

Edge enhancement

ABSTRACT

Computed Tomography (CT) imaging is one of the most widely-used and cost-effective technology for organ screening and diseases diagnosis. Because of existence of metallic implants in some patients, the CT images acquired from these patients are often corrupted by undesirable metal artifacts, which causes severe problem of metal artifact. Although there have been proposed many methods to reduce metal artifact, reduction is still challenging and inadequate, and results are suffering from symptom variance, second artifact and poor subjective evaluation. To address these problems, we propose a novel metal artifact reduction method based on generative adversarial networks to simultaneously reduce metal artifacts and enhance texture structure of corrected CT images. Specifically, we firstly incorporate interactive information (text) and imaging CT (image) into a comprehensive feature to yield multi-modal feature-fusion representation, which overcomes the representative ability limitation of single-modal data. The incorporation of interaction information constrains the feature generation to ensure symptom consistency between corrected and target CT. Then, we design an edge-enhance sub-network to avoid second artifact and suppress noise. Besides, we invite three professional physicians to evaluate corrected CT image subjectively. In this paper, We achieved average increment of 11.3% PSNR and 12.1% SSIM on DeepLesion dataset. The subjective evaluations by physicians show that ours outperforms over 6.3%, 7.1%, 5.50% and 6.9% in term of sharpness, resolution, invariance and acceptability, respectively. Our proposed method can achieve high-quality metal artifact reduction results.

© 2022 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

1. Introduction

Benefitting from the advantages in cost and imaging speed, CT has been a widely-used detection technology. CT images provide an efficient and effective way to medical diagnosis. Due to the existence of metallic implants, such as cochlear implants, prostheses, dental fillings and cardiac pace-maker, the corresponding CT images tend to be unacceptable in clinical diagnosis. The problem of metal artifact reduction (MAR) arises. In recent years, many correction algorithms are proposed to reduce the metal artifact, in-

cluding model-based [1–7] and deep learning-based MAR methods [8–15], and they have made promising breakthrough. Despite the progress made by existing algorithm-based MAR methods, there are still challenging and inadequate aspects as following. 1) all the existing MAR methods ignore the interaction information, i.e., initial information asked by the doctor before imaging. Generally, the doctor asks the patient about the illness firstly instead of imaging without any interaction. 2) metal artifacts are structured and non-local, but most of MAR algorithms are based on local and linear models. 3) there is few subjective evaluations about corrected CT images.

As shown in Fig. 1, professional assessments show that CNN-MAR [16] hallucinate high-frequency detail and edge so that they cannot conduct accurate diagnosis. Regardless of symptom variance in cGAN-CT [17] has proved that computer aided diagnosis algorithm incorporated with auxiliary prior information can gain bet-

[☆] These corresponding authors have the same contribution to this work and should be regarded as the first communication together.

* Corresponding authors.

E-mail addresses: hzwnet@swmu.edu.cn (Z. Huang), guo758881@foxmail.com (G. Zhang), linjz@cqupt.edu.cn (J. Lin), zhonglisha@swmu.edu.cn (L. Zhong).

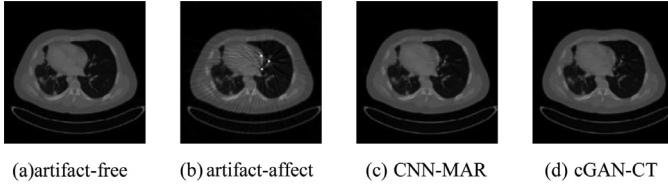


Fig. 1. The corrected results by two deep learning-based methods. (a) artifact-free (b) artifact-affect (c) CNN-MAR (d) cGAN-CT.

ter result than that without prior information. Although cGAN-CT embeds auxiliary label information to achieve satisfactory MAR results, this is practically inoperable. Because most of doctors provide diagnosis result or disease category based on CT images, but CT images are inaccessible before imaging. That is to say, the label is inaccessible and unknown before imaging. Then, most existing MAR methods regard the metal artifact as linear and local, and treat it as not the reduction but the restoration. Thus, corresponding MAR results may introduce second artifact. Finally, there is few subjective evaluations conducted on corrected CT images. Most MAR methods evaluate the corrected image use common way adopted in natural image evaluation. Different from natural image, medical image requires professional knowledge, but the evaluations of corrected image by existing MAR methods are still evaluated as same as the evaluations of realistic images.

Inspired by these three points, we develop a novel deep learning-based method which integrates multi-modal feature-fusion and edge-enhance to achieve high quality metal artifact reduction in CT images. The main contributions of this work can be summarized as follows:

- We propose multi-modal feature-fusion edge-enhance generative adversarial networks [18] (MFE-GAN) method for CT image metal artifact reduction, which represents the first effort that uses multi-modal feature to reduce metal artifact in metal artifact reduction. MFE-GAN simultaneously incorporates the deep and traditional handcrafted feature during artifact reduction.
- We incorporate multi-modal fusion feature into the proposed MFE-GAN to preserve symptom invariance. By considering the pre-interaction information and images simultaneously, the reconstructed CT images are close with the original images both in label and degree of illness evaluated by professional physician.
- The proposed MFE-GAN introduces the edge-enhance sub-network to maintain clear texture and detailed edges while effectively suppressing the noises. In particular, we design a sub-network branch to exact the edge and remove the noises produced by generative network through suppressing the noise-contaminated image regions. Then, our method fuses the exacted edge and denoised image to obtain the final corrected CT images.
- We conduct comprehensive experiments objectively and subjectively, including corrected CT images evaluation and diagnostic quality assessment. The extensive experimental results show that MFE-GAN has gained better performance than recent state-of-the-art MAR methods.

2. Methods

The deep structure of MFE-GAN, integrates generative, discriminative and edge-enhance networks, is shown in Fig. 2.

MFE-GAN generally consists of generative and discriminative network. The former network is used to reduce metal artifacts, and the latter network is designed to distinguish the authenticity between corrected and real CT images. The generative network is made up of two sub-networks: multi-modal feature-fusion sub-

network (MFSN) and edge-enhance sub-network (EESN). MFSN is used to generate intermediate corrected image where metal artifacts have been removed, the EESN is used to enhance edge and suppress noise.

2.1. Deep structure of MFE-GAN

Current GANs-based MAR algorithms always use two CNNs to learn the nonlinear mapping from artifact effected image domain to target image domain. But they hardly consider the symptom variance, second artifact and noise suppression. To deal with the limitation of deep learning-based MAR method, we fuse the multi-modal feature into correction and design an edge-enhance sub-network. Then, for better correction and training behavior, we adopt Wasserstein distance [19] to metric the distance between the distribution of corrected and real image, and add Spectral Normalization [20] behind activation layers to constrain discriminative performance of discriminative network. Thus, we have the following optimization problem:

$$\min_{\theta_g} \max_{\theta_d} L(\theta_g, \theta_d) = -E_{I^f \sim P_r} [\log D_{\theta_d}(I^f, t)] + E_{I^a \sim P_g} [\log D_{\theta_d}(G_{\theta_g}(I^a, t))] \quad (1)$$

where P_r and P_g denote the real and generative distribution, respectively. I^a is the pre-CT image which denotes the artifacts-affected CT, t denotes the interaction. The Spectral Normalization is used to normalize the weight matrix in discriminative network and it is not shown in Eq. (1). [21] has provide the detailed demonstration to show its contribution to controlling the Lipschitz constant.

2.2. Loss definition

To enforce the nonlinear mappings from artifact-affected image domain to target artifact-free image domain, we simultaneously introduce deep- and handcraft-feature loss functions to ensure that corrected CT images are consistent with real CT images both in low- and high- semantic level. The total loss incorporated in MFE-GAN can be denoted as:

$$L_{Total} = L_{Deep} + L_{Hand} \quad (2)$$

where L_{Deep} and L_{Hand} denote deep-feature and handcraft-feature loss, respectively. The former is designed to ensure high-level feature (such as context and semantic) consistency between corrected and real CT images, and the latter is utilized to ensure low-level feature consistency (such as edge, grayscale and brightness) between them. To be special, the deep-feature loss function can be represented as:

$$L_{Deep} = \alpha L_{CGen} + \beta L_{VGG/i,j} \quad (3)$$

where L_{CGen} denotes the conditional adversarial loss and $L_{VGG/i,j}$ is the visual perception loss [22] evaluated by VGG network [23]. α and β are hyper parameters used to control the weight between above losses. Moreover, the handcraft-feature loss function can be presented as:

$$L_{Hand} = \mu L_{pMSE} + \eta L_{TV} + \lambda L_{Edge} \quad (4)$$

where L_{pMSE} is pixel Mean Square Loss, L_{TV} is Total Variation loss, and L_{Edge} denotes the edge preservation loss. Similarly, μ , η and λ are also hyper-parameters that balance the weight proportion between above three losses. Followingly, we describe each item of each formulation in details.

2.3. Deep feature loss

The deep feature loss of this paper concludes three kinds of loss functions: conditional adversarial loss (L_{CGen}), content loss (L_{Con})

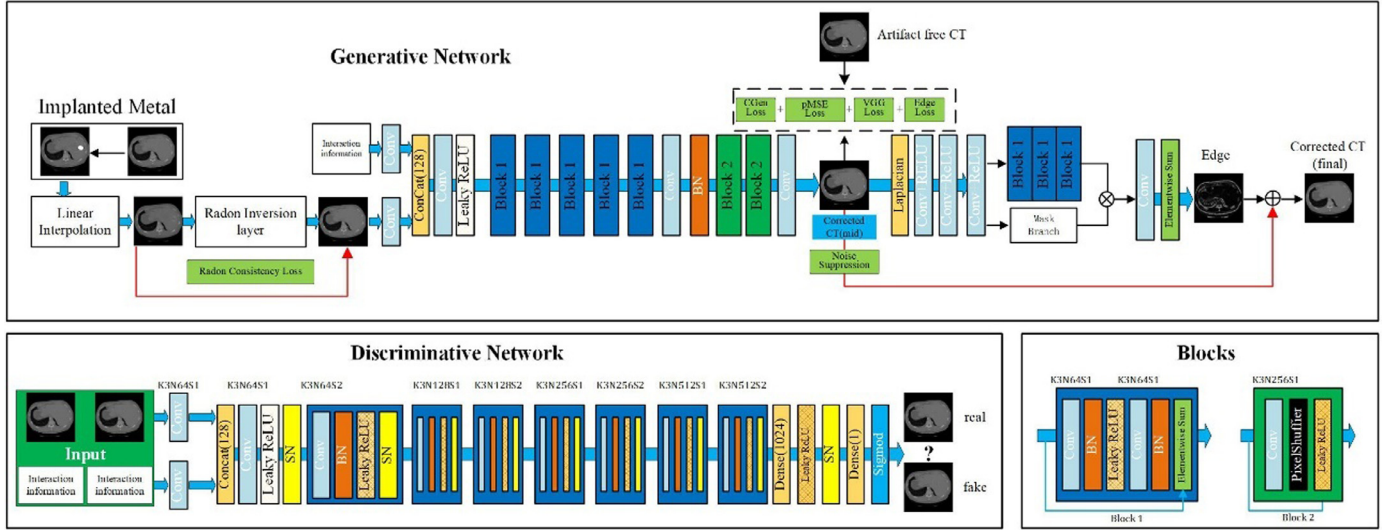


Fig. 2. The framework of proposed MFE-GAN.

and Radon consistency loss (L_{Rad}). The conditional adversarial loss is used to generate the image details that reside on the manifold of real CT image by fooling discriminative network. The visual perception loss is utilized to ensure high quality visual perception. As mentioned before, the visual perception loss is evaluated in VGG network [21]. Note that the VGG network in this paper is not trained on natural image dataset but DeepLesion dataset [24], which ensures that visual perception loss is based on medical image.

Owning to multi-modal feature-fusion of image and interactions information, L_{CGen} drives the generative network to generate the same pathological feature as original image, and it is defined as:

$$L_{CGen} = \sum_{n=1}^N \log D_{\theta_g}(G_{\theta_g}(I^{\alpha}, t)) \quad (5)$$

where t denotes the interactive information and N is the number of samples. As claimed and proved in previous work, using traditional handcraft-based feature, it can achieve high PSNR value, but the corrected CT image lacks high frequency details and tend to be over-smoothed. So, we introduce visual perceptual loss to preserve perceptual similarity, the visual perceptual loss function is defined as:

$$L_{VGG/i,j} = \frac{1}{W_{i,j}H_{i,j}} \sum_{i=1}^{W_{i,j}} \sum_{j=1}^{H_{i,j}} (\phi_{i,j}(I^f) - \phi_{i,j}(G_{\theta_g}(I^{\alpha}, t)))^2 \quad (6)$$

where $\phi_{i,j}$ indicates the feature map obtained by the j -th convolution (after activation) before the i th max-pooling within the VGG19 network. I^f and $G_{\theta_g}(I^{\alpha}, t)$ denote the reference artifact-free and corrected CT image, respectively. $W_{i,j}$ and $H_{i,j}$ represent the dimension of the respective feature maps within the VGG network.

As reported in previous works [25], the deep feature lacks interpretability, and it currently cannot replace the traditional handcrafted feature. So, apart from deep feature loss, we also embed the handcraft feature loss into the proposed MFE-GAN model.

2.4. Handcraft feature loss

There are increasing works that point out deep feature cannot be interpretable when exacting feature, and handcraft feature cannot be replaced by deep feature. Comparing with deep feature, handcraft feature is interpretable in exacting feature. So, apart

from those three deep feature losses, we also incorporate three handcraft feature losses into our proposed MFE-GAN model, including pixel Mean Square Loss L_{pMSE} , edge preservation loss L_{Edge} and Total Variation loss L_{TV} . L_{pMSE} is used to retain content invariance between corrected and real CT images in pixel-wise level. L_{Edge} is used to ensure the anatomical structure consistency between corrected and real CT images, and L_{TV} is used to suppress noise in corrected CT images. pMSE loss is one of the most widely used optimization target for image processing and transformation, and it can be represented as:

$$L_{pMSE} = \frac{1}{WH} \sum_{x=1}^W \sum_{y=1}^H (I_{x,y}^f - G_{\theta_g}(I_{x,y}^{\alpha}, t))^2 \quad (7)$$

where $I_{x,y}^{\alpha}$ is denoted as the tensor with the size of $W \times H \times C$. To preserve the anatomical structure consistency between corrected and real CT images, we define the edge preservation loss as:

$$L_{Edge} = \sum_{i=1}^c V_c[G_{\theta_g}(x, t)]^T M_{lf} V_c[G_{\theta_g}(x|l, z)]^T \quad (8)$$

where M_{lf} is Matting Laplacian [26], which represents standard linear system and only depends on the inputs to ensure anatomical structure to be same with real anatomical structure during prediction. $V_c[\cdot]$ denotes vectorized corrected CT image in channel c . Both artifact-free and corrected CT will be transformed in the format of RGB.

It is well known that the generative network in GANs always samples from low-dimension noise, then the generated MAR results may contain additional details (e.g., second artifacts) beyond original input and even with noise. To further avoid the second artifacts and noise meanwhile enhancing the edge, we design another sub-network to achieve these goals, in which total variation loss is incorporated.

2.5. Noise suppression and edge enhancement

The edge-enhance sub-network (EESN) firstly extracts the edge of intermediate corrected CT image output by multi-modal feature-fusion sub-network (MFSN), then it uses TV loss to suppress additional details (e.g., noise and second artifact) of corrected image. Finally, noiseless corrected image and enhanced edge are merged into final corrected CT image.

As shown in Fig. 2, the EESN module takes the stage results I_{mid}^c obtained from MFSN module as input and outputs desired corrected CT image. In this paper, we use Laplacian operator [27, 28]

to highlight and detect regions of rapid intensity change to extract the edge.

The Laplacian operator $L(x, y)$ of an image $I(x, y)$ is always defined as its second derivative. Laplacian operator possesses isotropy and rotation invariance, which makes the image produce a steep zero-crossing point at the edge after the second derivatives. For better computation and corrected results, we utilize the 3×3 discrete convolution kernel $([-1, -1, -1], [-1, 8, -1], [-1, -1, -1])$ as Laplacian operator. Then, extracted edge $E(x, y)$ can be represented as:

$$E(x, y) = L(x, y) \otimes I(x, y) \lim_{x \rightarrow \infty} \quad (9)$$

where $\otimes$ represents pixel-wise convolution operation, and $L(x, y)$ is Laplacian operator which can be defined as:

$$L(x, y) = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2} \quad (10)$$

For better performance, we utilize the ResNet block [23] to extract edge maps and transform them into low-resolution space simultaneously, which makes significant contribution to computing. We also use a mask branch to learn the image mask to detect. Subsequently, the enhanced edge maps are projected onto high-resolution space by a sub-pixel convolution operation. The above-mentioned operations can be formulated as follows:

$$I_{edge}^c = PS(E(B(I_{mid}^c)) \otimes M(B(I_{mid}^c))) \quad (11)$$

where $B(\bullet)$ is the bi-cubic downsampling operator and $E(\bullet)$ denotes the dense sub-network in edge enhance part used for feature extraction and fusion. $M(\bullet)$ represents the mask branch to learn the image mask to suppress the noise and the false edges. $PS(\bullet)$ is the up-sampling operation with the sub-pixel convolution to transform the edge maps into high-resolution space. I_{mid}^c is the intermediate corrected CT image outputted by MFSN part, and it is not the final corrected CT image.

Considering the introduction of additional details in intermediate corrected CT image beyond artifact-free CT image, we use TV loss to reduce the noise or undesired details in I_{mid}^c . The TV loss is another widely used optimization target for image processing, which has been demonstrated to be state-of-the-art in reducing noise in piecewise-constant images. In this paper we formulate TV-based optimization as:

$$I_{base}^c = \min \left[\sum_{i=1}^{W-1} \sum_{j=1}^{H-1} ((I_{mid}^c)^{i,j+1} - (I_{mid}^c)^{i,j})^2 + \sum_{i=1}^{W-1} \sum_{j=1}^{H-1} ((I_{mid}^c)^{i+1,j} - (I_{mid}^c)^{i,j})^2 \right] \quad (12)$$

where I_{mid}^c denotes the corrected CT image without noise suppression.

The final denoised corrected CT image can be formulated as:

$$I^c = I_{base}^c \oplus I_{edge}^c \quad (13)$$

where I_{base}^c is noiseless and suppressed intermediate corrected CT image and I_{edge}^c is enhanced edge. $\oplus$ is pixel-wise addition and I^c denotes the final corrected CT images.

2.6. Algorithm and optimization

Together with the all the loss above, the total loss function of proposed algorithm are summarized as Eq. (2), which consists of deep feature loss and handcraft feature loss.

The deep feature loss contains conditional adversarial loss L_{CGen} and perception visual loss $L_{VGG/i,j}$. The former drives generative network to generate the details that reside the manifold of artifact-free CT images and ensure pathological invariance, the latter is

used to take the perceptual similarity into account and yield the MAR results which is in line with the human's perception vision.

It is claimed that traditional handcrafted features cannot be replaced by depth features despite depth features achieve good results. Thus, we also introduce three handcrafted features, including pixel Mean Square Error L_{pMSE} , edge preservation loss L_{edge} and total variation loss L_{TV} . Pixel MSE loss which is define on the pixel-wise image difference can be used as the optimization cost function for the fast MAR. Considering that edge and texture are always concerned in medical image and clinical diagnosis, we propose edge preservation loss to preserve edge and texture in corrected CT images. Besides, generative network introduces noise or undesired details into corrected CT image, we incorporate TV loss to suppress and avoid the additional details beyond artifact-free CT images.

It is easy to demonstrate that the total objective function Eq. (2) is non-convex, so we adopt the mini-batch stochastic gradient descent method and alternating learning strategy to train the generative and discriminative network following previous deep learning algorithms.

As shown in Fig. 2, the generative network of proposed MFE-GAN model consists of two sub-networks, i.e., MFSN and EESN. In practice, for training convenience and better MAR results, we firstly pre-train each sub-network until it converges, and fine-tune the whole generative network. Once the generative network for MAR has been trained based on Total loss, we can apply it to any artifacts-affected image inputs and it will result in the reference artifacts-free images. The application can be simplified as $G(I^a) = I^c$, where I^a is artifact-affected CT image and I^c denoted the metal artifact reduction CT images which is very close to artifact-free CT image.

$$\nabla_{\theta_d} \frac{1}{n} \sum_{k=1}^n - \left[\log D_{\theta_d}(I^f | t)^k + \log(D_{\theta_d}(G_{\theta_d}(I^a | t)^k)) \right] \quad (14)$$

$$\begin{aligned} \nabla_{\theta_g} \frac{1}{n} \sum_{k=1}^n & \left[\alpha \log D_{\theta_d}(G_{\theta_g}(I^a | t)^k) + \frac{\beta}{W_{i,j} H_{i,j}} \sum_{x=1}^{W_{i,j}} \sum_{y=1}^{H_{i,j}} (\phi_{i,j}((I^f)^k) \right. \\ & - \phi_{i,j}(G_{\theta_g}(I^a | t)^k))^2 + \frac{\mu}{WH} \sum_{x=1}^W \sum_{y=1}^H (I_{x,y}^f - G_{\theta_g}(I_{x,y}^a | t)^k)^2 \\ & + \frac{\eta}{WH} \sum_{i=1}^{W-1} \sum_{j=1}^{H-1} ((G_{\theta_g}(I^a | t))^{i,j+1} - (G_{\theta_g}(I^a | t))^{i,j})^2 \\ & + ((G_{\theta_g}(I^a | t))^{i+1,j} - (G_{\theta_g}(I^a | t))^{i,j})^2 \\ & \left. + \lambda \sum_{i=1}^c V_c[G_{\theta_g}(y | l, z)]^T M_{MRI} V_c[G_{\theta_g}(y | l, z)]^T \right] \end{aligned} \quad (15)$$

3. Experiment and analysis

3.1. Datasets and implementation details

Dataset. We firstly use a large-scale CT dataset named DeepLesion [24] to simulate the metal artifacts and reduction. DeepLesion is established for lesion detection, and contains over 30,000 CT images collected from over 4000 patients and matched annotated information. We choose two sub-sets of DeepLesion which consists over 8000 images to validate the deep learning-based MAR methods due to its high diversity and quality. We use 6000 CT images and the corresponding annotated information for training set, and the rest for testing. Then, we collect 50 metal affected CT images from Southwest Medical University Affiliated Hospital and the interaction information before imaging to test the performance of

Table 1

IFC, PSNR, FSIM, SSIM and MSIM values (mean $\pm$ stds) on DeepLesion dataset. Bold and underlined indicate the best and second performance, respectively.

Method	IFC	PSNR	SSIM	MSIM	FSIM
LI [1]	1.93 $\pm$ 0.11	26.44 $\pm$ 1.28	0.883 $\pm$ 0.19	0.967 $\pm$ 0.21	0.912 $\pm$ 0.28
NMAR [8]	2.04 $\pm$ 0.14	27.67 $\pm$ 0.52	0.896 $\pm$ 0.11	0.971 $\pm$ 0.18	0.928 $\pm$ 0.17
CNN-MAR [16]	2.07 $\pm$ 1.23	28.33 $\pm$ 0.47	0.914 $\pm$ 0.10	0.983 $\pm$ 0.12	0.931 $\pm$ 0.16
cGAN-CT [17]	2.18 $\pm$ 0.14	29.55 $\pm$ 0.66	0.928 $\pm$ 0.14	0.987 $\pm$ 0.11	0.931 $\pm$ 0.13
DuDoNet [14]	2.34 $\pm$ 0.10	30.88 $\pm$ 0.65	0.934 $\pm$ 0.09	0.990 $\pm$ 0.12	0.942 $\pm$ 0.10
MFE-GAN	2.35 $\pm$ 0.10	31.25 $\pm$ 0.58	0.939 $\pm$ 0.09	0.990 $\pm$ 0.14	0.953 $\pm$ 0.12

trained model. These clinical artifact-affect CT images, called CLD, are regarded as out-of-the-samples to further test the performance of proposed method.

Evaluations. We compare our method with five state-of-the-art MAR method, including LI [1], NMAR [8], CNN-MAR [16], cGAN-CT [17] and DuDoNet [14]. We firstly evaluate the correct CT images with several evaluation methods including PSNR, IFC [29], SSIM, MSIM [30] and FSIM [31]. Since the texture structure draws more attention in the clinical, we mainly concern the last three indexes. Then, we adopt Sliced Wasserstein distance (SWD) [32], Inception Score (IS) [33], Frechet Inception Distance (FID) [34], GAN-train and GAN-test [35] to evaluate network performance of deep learning-based MAR approaches. More importantly, we invite three radiologist who have mean clinical experience over 5 years to conduct assessment of diagnostic quality, which helps to verify the pathological and degree invariance.

Details. We implement our model using the Tensorflow framework. All the experiments are conducted on Intel(R) Xeon(R) Gold6148CPU with 20 cores and eight Tesla V100-SXM2GPUs using same setting to ensure impartiality and objectivity. To train the model, we use the Adam [36] optimizer with $(\beta_1, \beta_2) = (0.9, 0.99)$. According to the prior knowledge provided in [37–40], we set the initial learning be 10^{-4} . Considering the limited hardware, the batch size is set to 16. For better corrected results, the iteration is fixed to 9×10^5 . As displayed in Eq. (3) and 4, the total loss of MFE-GAN involves six hyper-parameters. So, in order to avoid overfitting, we directly set $\mu = 0.95$ since it is used to ensure content invariance and it accounts most of weight. Then, we fix $\eta = 10^{-8}$ because slightly bigger weight of TV loss will cause over-smoothing. Analogously, we set $\lambda = 5 \times 10^{-6}$ since it cannot be set to too large or it will cause over-sharpening as claimed in [4]. Then, the rest hyper-parameters are set automatically in the range of $[10^{-8}, 10^2]$ with a multiplication step of 10 according to the results of cross-validation.

We add the metal mask to artifact-free CT image to simulate artifact-affect CT image using the same mechanism adopted in [15]. To be special, we utilize the 100 metal shapes provided by [15]. Then, we pair 90 metal shapes with 4000 images, which result in 360,000 combinations in training set. Sequentially, we pair rest 10 metal shapes with 200 images, which yield 2000 combinations in testing set. Both metal shape and image will be resized to 256×256 .

3.2. Evaluations on simulation data

We evaluate our MFE-GAN method with mentioned five MAR methods on DeepLesion dataset. The visual results are shown in Fig. 3 and the quantitative comparisons are shown in Table 1.

Fig. 3 generally shows that the deep learning-based MAR methods (i.e., CNN-MAR, cGAN-CT, DuDoNet and MFE-GAN) outperform the MAR methods (i.e., LI and NMAR). And all deep learning-based methods yield seemingly compelling MAR results. However, some information (e.g., texture) is lost when zooming the regions of interest (ROI) as shown in Fig. 3. Visually, cGAN-CT, DuDoNet and

Table 2

SWD, IS, FID, GAN-train and GAN-test values on DeepLesion dataset. SWD and FID: the lower is better. IS, GAN-train and GAN-test: the higher is better.

Method	SWD	IS	FID	GAN-train	GAN-test
L1 [1]	5.34	21.55	9.81	0.74	0.69
CNN-MAR [16]	4.88	24.35	9.07	0.81	0.79
cGAN-CT [17]	4.47	25.00	8.74	0.85	0.79
DuDoNet [14]	3.79	28.86	7.43	0.91	0.82
MFE-GAN	3.08	30.17	6.98	0.91	0.87

our proposed MFE-GAN gain better MAR results than others according to the observation on ROI.

For better comparison, we also use some objective indexes to conduct comprehensive evaluations, including IFC, PSNR, SSIM, MSIM and FSIM. The first two are related with perception vision, and the last three are related with texture structure. The results are shown in Table 1. We can also conclude that deep learning-based MAR methods outperform the MAR methods. Then, owing to introduction of edge-enhance sub-network, ours achieves best performance comparing with other deep learning-based comparisons. MFE-GAN achieves average increment of 7.8% PSNR, 5.3% IFC, 10.2% SSIM, 6.7% MSIM and 6.7% FSIM on DeepLesion.

It can be observed that our proposed MFE-GAN does not reach highest score in each index. Specially, MFE-GAN incurs decrement of 4.3% PSNR and 5.6% IFC comparing with DuDoNet on DeepLesion dataset. DuDoNet takes sinogram and image domain into account, and results in satisfactory MAR results in human's perception. As mentioned before, the texture draws more attention compared to perception. The indexes related to edge and texture, i.e., SSIM, MSIM and FSIM, show that our method handles medical image better. Comparing with DuDoNet, the decrement by our method is minus.

Apart from the evaluation of MAR results, we have also conducted some experiments on network performance of deep learning-based MAR methods. The results are shown in Table 2.

Table 2 shows that DuDoNet gains the highest FID and IS score, and ours gains the SWD, GAN-train and GAN-test scores. It has been proved that FID and IS are built on Inception network which is trained on ImageNet dataset. That is to say, they are suitable for evaluating natural images. Thus, combining the results in Fig. 3 and Tables 1 and 2, we can conclude that our proposed MFE-GAN can yield better MAR CT image in edge and texture preservation which are always concerned in medical image and clinical diagnosis.

3.3. Evaluations on clinical data

Apart from the results evaluated on simulation image by adding metal mask to real image in DeepLesion, we have also conducted some evaluation on clinical samples collected in the hospital. The visual MAR results are shown in Fig. 4.

To highlight corrected region, we have done some local marking in the region of interest. From Fig. 4, we can see that our method and DuDoNet outperform others and achieve comparable performance in clinical artifacts reduction. Considering that clinical sam-

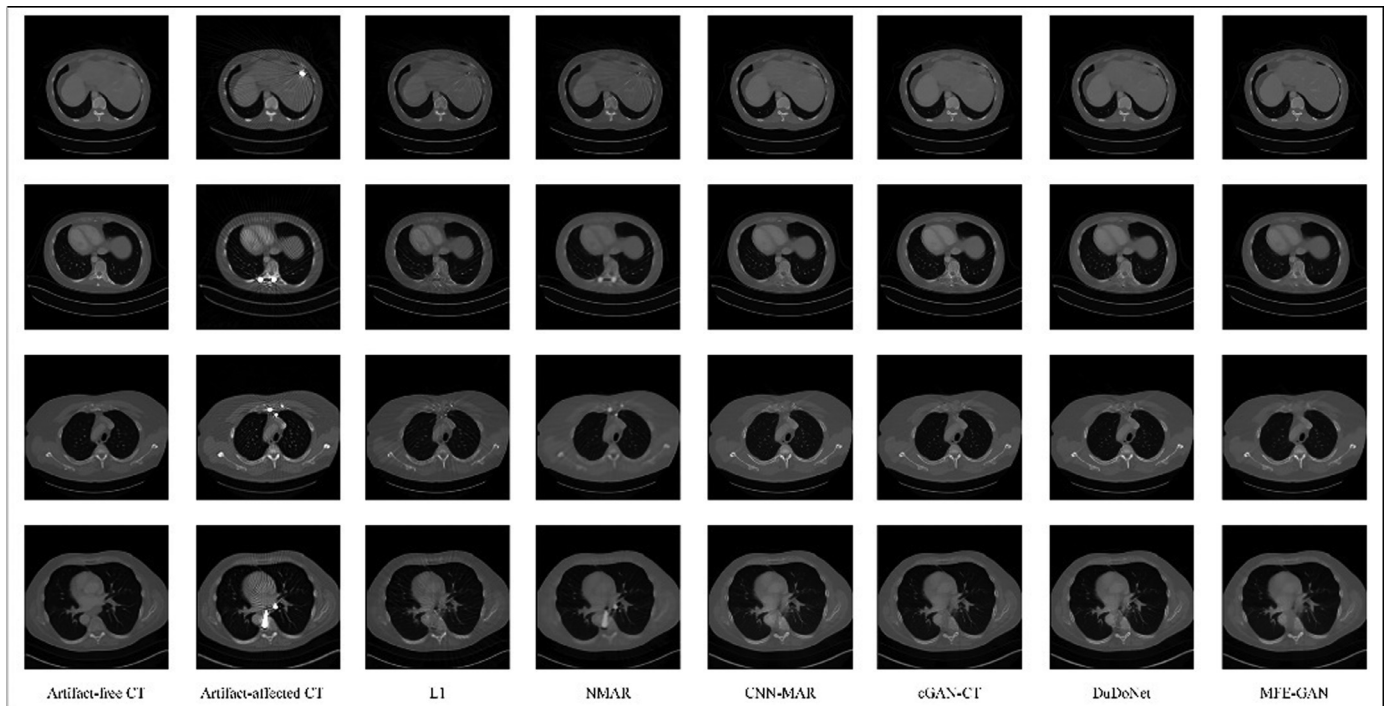


Fig. 3. The visual MAR results on DeepLesion dataset.

Table 3

Diagnostic Quality Assessment in Terms of Subjective Quality Scores for Different Algorithms (Mean $\pm$ STDS).

Methods	DeepLesion				CLD			
	sharpness	resolution	invariance	acceptability	sharpness	resolution	invariance	acceptability
LI [1]	1.3 $\pm$ 0.21	1.5 $\pm$ 0.25	0.1 $\pm$ 0.09	1.3 $\pm$ 0.21	1.2 $\pm$ 0.16	1.4 $\pm$ 0.24	0.0 $\pm$ 0.00	1.2 $\pm$ 0.16
NMAR [8]	1.6 $\pm$ 0.24	1.9 $\pm$ 0.49	0.2 $\pm$ 0.16	1.7 $\pm$ 0.41	1.4 $\pm$ 0.24	1.6 $\pm$ 0.24	0.1 $\pm$ 0.09	1.6 $\pm$ 0.24
cGAN-CT [17]	3.2 $\pm$ 0.16	3.3 $\pm$ 0.21	0.5 $\pm$ 0.25	3.4 $\pm$ 0.24	3.1 $\pm$ 0.19	3.2 $\pm$ 0.16	0.4 $\pm$ 0.24	3.3 $\pm$ 0.24
CNN-MAR [16]	2.0 $\pm$ 0.24	2.3 $\pm$ 0.21	0.2 $\pm$ 0.16	1.7 $\pm$ 0.41	1.5 $\pm$ 0.25	1.9 $\pm$ 0.29	0.2 $\pm$ 0.16	1.5 $\pm$ 0.25
DuDoNet [14]	3.7 $\pm$ 0.16	3.5 $\pm$ 0.16	0.5 $\pm$ 0.25	3.8 $\pm$ 0.21	3.4 $\pm$ 0.16	3.6 $\pm$ 0.24	0.4 $\pm$ 0.24	3.7 $\pm$ 0.21
MFE-GAN	3.9 $\pm$ 0.24	3.8 $\pm$ 0.24	0.7 $\pm$ 0.16	4.0 $\pm$ 0.24	3.7 $\pm$ 0.24	3.6 $\pm$ 0.21	0.7 $\pm$ 0.16	3.9 $\pm$ 0.24

ples have no references and evaluation needs more expertise. So, we invite three chief physicians who possess over 5 years clinical experience in radiology to conduct image analysis independently using the standards in SNSRGAN [25], including sharpness, resolution, invariance and acceptability. We randomly construct 10 sets of testing samples (5 sets for DeepLesion and 5 sets for CLD), and each set is consisted of 10 artifact-corrected CT images. The mean scores with standard deviation by chief physicians are shown in Table 3.

As shown in Table 3, the proposed MFE-GAN achieves best diagnostic quality assessment especially in sharpness and invariance, which owns to the introduction of edge-enhanced sub-network and interactive information. Comparing with other MAR methods, our method gains average increment of 1.54, 1.30, 0.40 and 1.62 in sharpness, resolution, invariance and acceptability in DeepLesion dataset, respectively. A similar trend can be found in CLD dataset, that is our methods outperforms average 1.58, 1.26, 0.48 and 1.64 in corresponding index comparing with other MAR methods.

3.4. Effect of super-resolution

Combing the MAR results on simulation and clinical dataset, we can see that our method gains better MAR results comparing with other methods. Even though, the visual perception is worse than original artifact-free CT image and the diagnostic quality assessments especially in ‘acceptability’ are slightly lower. In other

words, all the compared method currently cannot reach the level of clinical application.

In order to improve corrected CT image, we use a simple super-resolution post-processing to further improve resolution and details of corrected CT images. In practice, we use a simple super-resolution network SNSRGAN [25] to achieve this goal. The SNSRGAN is trained on another 8 sub-sets in DeepLesion dataset, and these sub-sets are different from the 2 sub-sets which are used to trained MFE-GAN. Then, we apply trained SNSRGAN to corrected CT images output by proposed MFE-GAN. The improved results are shown in Figs. 5 and 6.

The comparisons in Fig. 5 show that this simple post-processing make the detail and resolution of corrected CT image be better. From Fig. 6 we can observe that the post-processing operation contribute significantly in sharpness and resolution. Specially, SR corrected CT image achieves increment of 0.7 sharpness and 0.6 resolution both on DeepLesion and CLD dataset. All these results indicate that it can further improve MAR results and shorten gap between theoretical research and clinical applications.

4. Discussion

The proposed MFE-GAN can achieve high-quality metal artifact reduction results owing to two important components, that is: (1) MFE-GAN fuses the interaction information and image to constrain the feature generation and (2) MFE-GAN embeds the edge-enhance

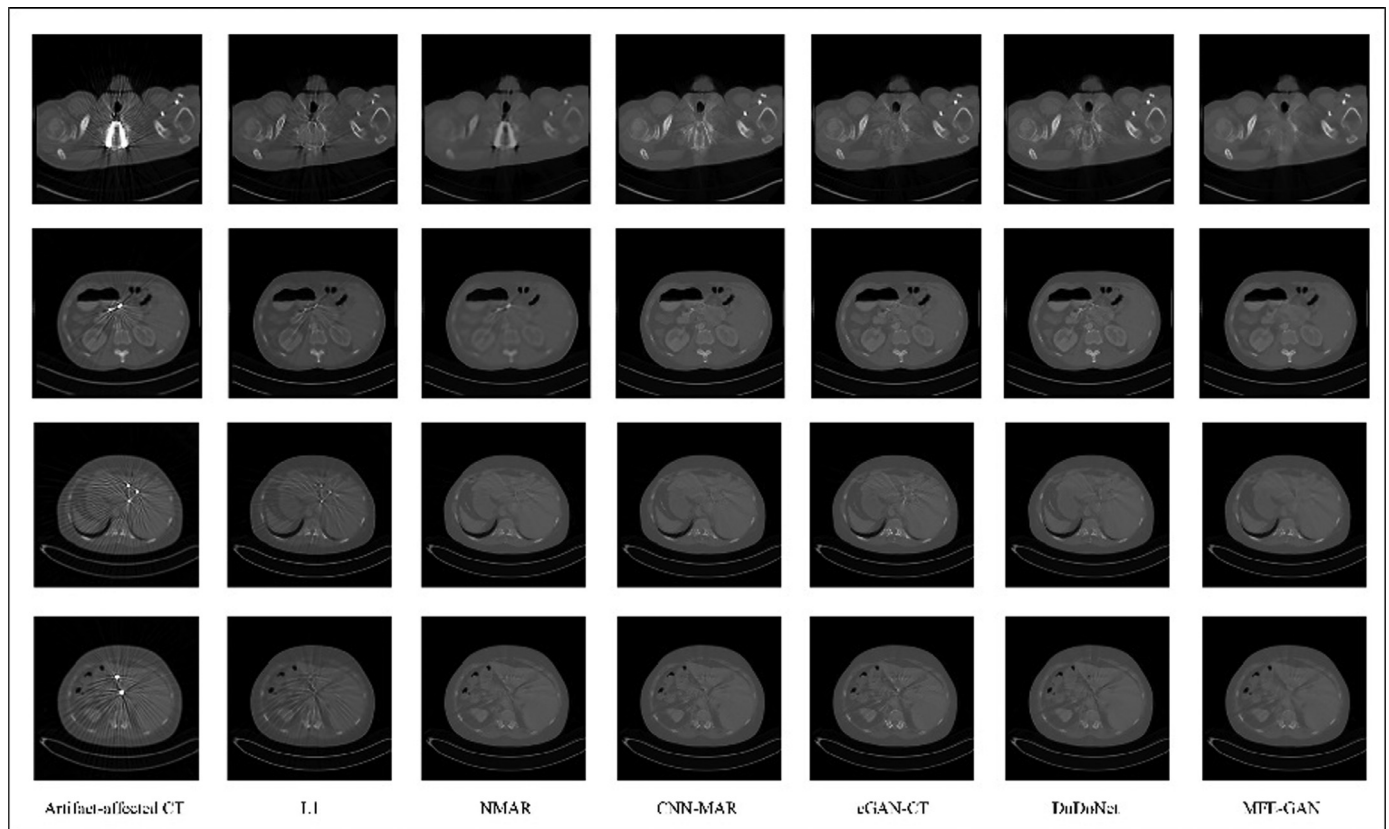


Fig. 4. The visual MAR results on CLD dataset.

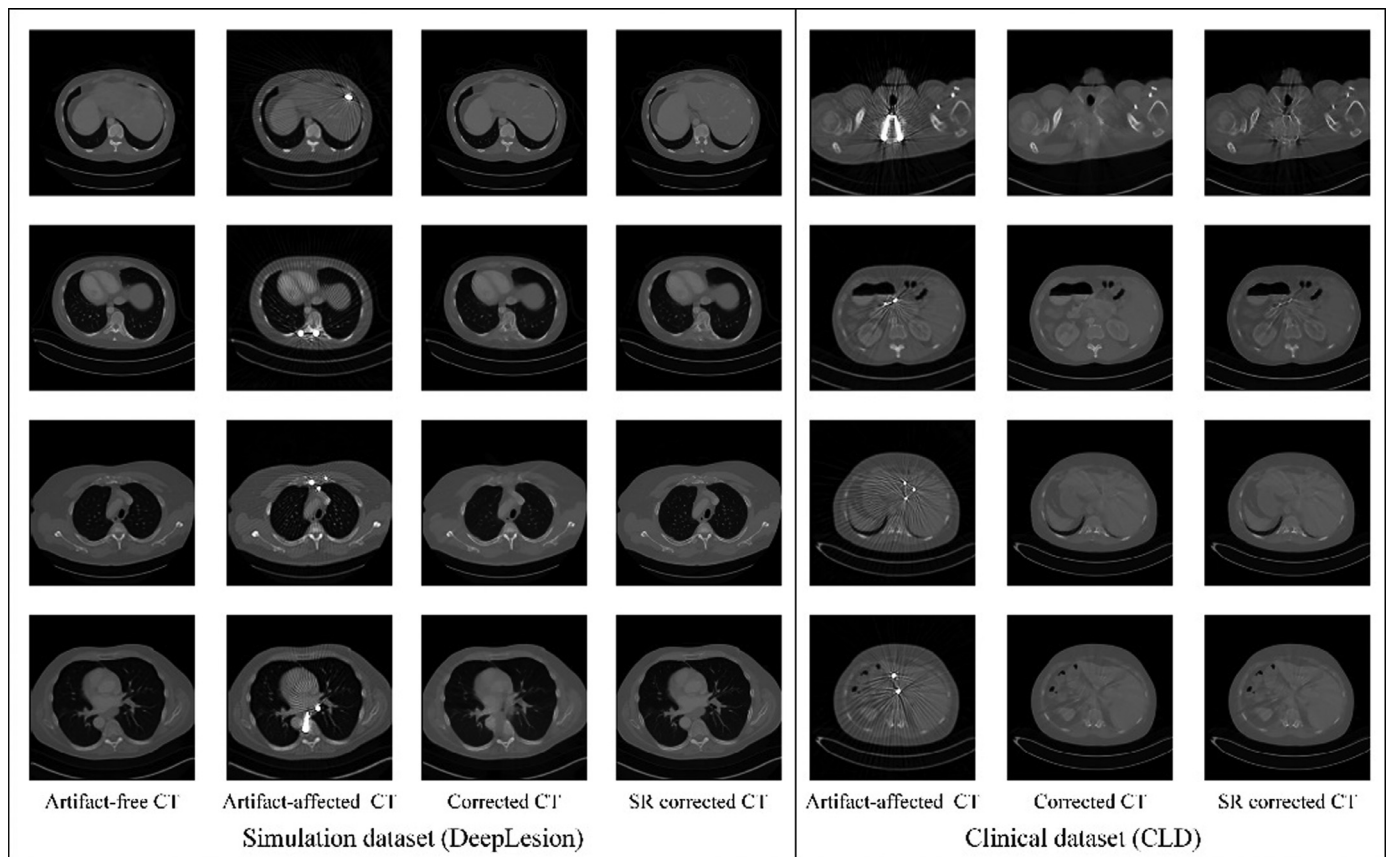
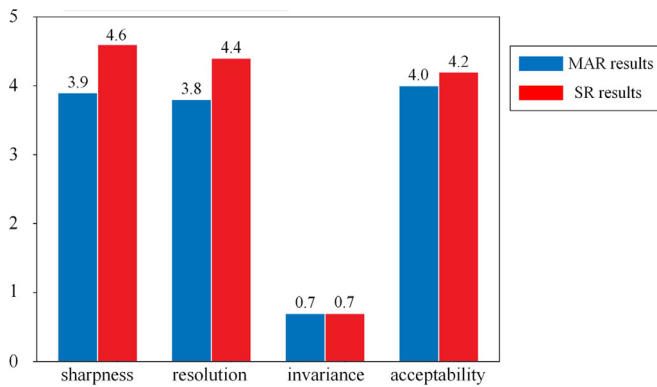
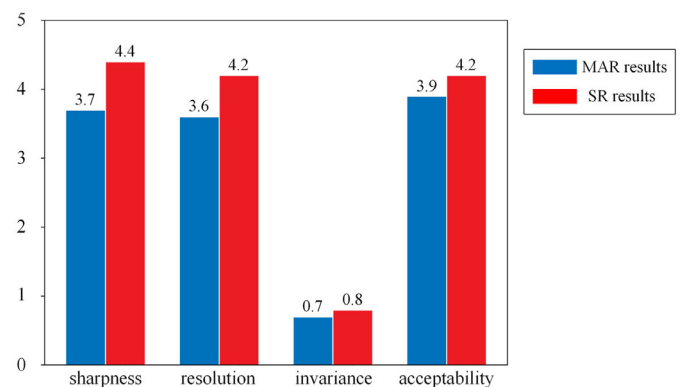


Fig. 5. The visual comparisons on corrected CT image and corresponding SR results. First row of images is artifact-affected CT, the second is corrected CT and the rest are SR corrected CT images.



(a) The Assessment on DeepLesion



(b) The Assessment on CLD

Fig. 6. The diagnostic quality assessment comparison on corrected CT image and corresponding SR result. (a) The Assessment on DeepLesion (b) The Assessment on CLD.**Table 4**

IFC, PSNR, FSIM, SSIM and MSIM values (mean $\pm$ stds) of MFE-GAN and its variants on DeepLesion dataset.

Method	IFC	PSNR	SSIM	MSIM	FSIM
MFE-GAN-I	2.147 $\pm$ 0.11	30.54 $\pm$ 0.16	0.925 $\pm$ 0.03	0.983 $\pm$ 0.03	0.941 $\pm$ 0.08
MFE-GAN-S	2.122 $\pm$ 0.14	28.88 $\pm$ 0.26	0.887 $\pm$ 0.05	0.944 $\pm$ 0.04	0.909 $\pm$ 0.07
MFE-GAN-V	1.975 $\pm$ 0.21	26.15 $\pm$ 0.17	0.901 $\pm$ 0.10	0.956 $\pm$ 0.12	0.918 $\pm$ 0.16
MFE-GAN-P	1.993 $\pm$ 0.10	26.42 $\pm$ 0.18	0.919 $\pm$ 0.12	0.978 $\pm$ 0.13	0.935 $\pm$ 0.09
MFE-GAN	2.350 $\pm$ 0.17	31.25 $\pm$ 0.28	0.939 $\pm$ 0.06	0.990 $\pm$ 0.11	0.953 $\pm$ 0.09

sub-network, which simultaneously enhances the edge and avoid second artifact or noise. To further analyze the function of each component in MFE-GAN, we investigate several variants of MFE-GAN as follows, and the comparisons on DeepLesion dataset are summarized in Table 4.

- MFE-GAN-I: The MFE-GAN variants without using interaction information;
- MFE-GAN-S: The MFE-GAN variants without using edge-enhance sub-network;
- MFE-GAN-V: The MFE-GAN variants without using VGG loss;
- MFE-GAN-P: The MFE-GAN variants without using pMSE loss;

The usage of multi-modal feature-fusion has received much approval in medical image processing because of better performance comparing with single-modal feature. Based on this point, we fuse the neglected but accessible interactive information and medical image to gain better MAR results, which is the first effort in the field of metal artifact reduction. Then, considering that GANs may generate additional details (e.g., noise or second artifact) in corrected CT image beyond original image, we design an edge-enhanced sub-network to suppress and eliminate these generated additional details, which is another key point. It can be also seen that MFE-GAN has made much contribution in preserving texture structure and edge by introducing edge-enhance sub-network. Besides, The VGG loss is designed for better perception vision and the pMSE loss is used to ensure content invariance. Compared with other variances, MFE-GAN has achieved average increments of 14.3% IFC, 12.1% PSNR, 3.4% SSIM, 2.6% MSIM and 3.0% FSIM. We choose the Laplacian operator with 3×3 discrete convolution kernel to exact edge of intermediate corrected CT image. We have compared several edge exaction operator and different sizes of kernel. The comparisons are shown in Table 5 and 6.

As shown in Table 5, we compare five edge exaction operators and the results show that the Laplacian and Candy operator can obtain higher MAR performance than others. Roberts, Sobel and Prewitt operator involve both the edge and its neighbor pixel. Since the intermediate corrected CT image contains additional details as

Table 5

The objective evaluation values of edge detection operators on DeepLesion dataset.

Operator	PSNR	IFC	SSIM	MSIM	FSIM	GAN-train	GAN-test
Roberts	30.21	2.23	0.931	0.985	0.950	0.89%	84.7%
Sobel	29.90	2.16	0.932	0.984	0.948	0.85%	82.2%
Candy	29.55	2.04	0.930	0.982	0.940	0.84%	82.1%
Prewitt	29.50	2.07	0.929	0.978	0.934	0.84%	81.4%
Laplacian	31.25	2.35	0.939	0.990	0.953	0.91%	87.0%

Table 6

The objective evaluation values of Laplacian operators with different size on DeepLesion dataset.

Size	PSNR	IFC	SSIM	MSIM	FSIM	GAN-train	GAN-test
3×3	31.25	2.35	0.939	0.990	0.953	0.91%	87.0%
5×5	31.49	2.66	0.942	0.992	0.958	0.91%	87.2%
7×7	31.95	2.74	0.940	0.994	0.963	0.92%	87.4%
9×9	31.69	2.70	0.941	0.992	0.967	0.91%	87.2%

mentioned before, then the exacted edge contains real edge and other information. In contrast, the Laplacian operator only focuses on edge and not the other parts. Thus, the final merged corrected CT image contains less additional information and tend to be more satisfactory. Comparing with Laplacian operator, the candy operator achieves higher evaluations because it involves multiple operations, including smoothing, filtering, exacting and enhancing. Considering comparable performance and computation cost, we use Laplacian instead of candy to exact edge.

Table 6 shows that using Laplacian operator with the size of 7×7 achieves slightly better performance than others. The larger is the size of kernel, the more neighbor information is considered. But oversized kernel will reduce computation performance. As described before, we consider both performance and computation, i.e., when using Laplacian operator with the size of 3×3 , 5×5 and 7×7 achieves comparable MAR results but the first size costs the least training time. Thus, in practice we choose the Laplacian operator with the size of 3×3 to exact edge.

5. Conclusion

In this paper, we present a novel GANs-based metal artifact reduction method, which addresses the limitation of single-modal feature and the problem of second artifact and symptom variance. The proposed method concludes MFSN and EESN. MFSN fuses the interactive information and CT image for multi-modal feature with strong representation, which ensures the symptom consistency between corrected and artifact-free CT image. EESN is designed to

preserve the edge and texture structure after suppressing the additional details generated by GANs, which provides better MAR results. To distinguish from the evaluations of natural image and conduct comprehensive evaluations, professional chief physician has been used to assess the performance of our proposed method. The extensive experiments show that our method outperforms the state-of-the-art MAR approaches. In the future, we will combine multiple accessible source (e.g., text, image and sinogram) into a unified framework to achieve artifact reduction caused by multiple implanted materials.

Declaration of Competing Interest

The authors declare no conflicts of interest.

Ethical compliance

There is no ethics approval required for this paper.

Acknowledgment

The project was funded by Central Nervous System Drug Key Laboratory of Sichuan Province, Medicine & Engineering & Informatics Fusion and Transformation Key Laboratory of Luzhou City, the National Natural Science Foundation of China (Grant no. 61971079, 61801067), Science and Technology Department of Sichuan Province (2020YFQ0025, 2020YJ0151), Key Research Project of Southwest Medical University (2021ZKZD019).

References

- [1] W. Kalender, R. Hebel, J. Ebersberger, Reduction of CT artifacts caused by metallic implants, *Radiology* 164 (2) (1987) 576–577.
- [2] M. Bal, L. Spies, Metal artifact reduction in CT using tissue-class modeling and adaptive prefiltering, *Med. Phys.* 33 (8) (2006) 2852–2859.
- [3] D. Prell, Y. Kyriakou, M. Beister, W.A. Kalender, A novel forward projection-based metal artifact reduction method for flat-detector computed tomography, *Phys. Med. Biol.* 54 (21) (2009) 6575–6591.
- [4] G. Wang, D. Snyder, Iterative deblurring for CT metal artifact reduction, *IEEE Trans. Med. Imaging* 15 (5) (1996) 657–664.
- [5] D. Man, J. Nuyts, P. Dupont, G. Marchal, P. Suetens, An iterative maximum-likelihood polychromatic algorithm for CT, *IEEE Trans. Med. Imaging* (2022).
- [6] C. Lemmens, D. Faul, J. Nuyts, Suppression of metal artifacts in CT using a reconstruction procedure that combines MAP and projection completion, *IEEE Trans. Med. Imaging* 28 (2) (2009) 250–260.
- [7] M. Bal, H. Celik, K. Subramanyan, E. Kai, L. Spies, A radial adaptive filter for metal artifact reduction, *SPIE Med. Imaging* (2005) 2075–2082.
- [8] E. Meyer, R. Raupach, M. Lell, B. Schmidt, M. Kachelrie, Normalized metal artifact reduction (NMAR) in computed tomography, *Med. Phys.* 37 (10) (2010) 5482–5493.
- [9] H. Zhang, B. Dong, B. Liu, A reweighted joint spatial-radon domain CT image reconstruction model for metal artifact reduction, *SIAM J. Imaging Sci.* 11 (1) (2018) 707–733.
- [10] H. Sungsoo, M. KlausGenerative, Mask pyramid network for CT/CBCT metal artifact reduction with joint projection-sinogram correction, *SPIE Med. Imaging* (2016) 833–841.
- [11] L. Gjesteb, Q. Yang, X. Yan, Z. Ye, W. Ge, Deep learning methods to guide CT image reconstruction and reduce metal artifacts, *SPIE Med. Imaging* (2017) 21–27.
- [12] H. Park, Y. Chung, S. Lee, H. Kim, J. Seo, Sinogram-consistency learning in CT for metal artifact reduction, *SPIE Med. Imaging* (2017) <https://arxiv.org/abs/1708.00607>.
- [13] X. Shiyu, D. Hao, Deep residual learning enabled metal artifact reduction in CT, *Phys. Med. Imaging* (2018) 1057330.
- [14] W. Lin, H. Liao, C. Peng, X. Sun, J. Zhang, J. Luo, R. Chellappa, S.K. Zhou, DuDoNet: dual domain network for CT metal artifact reduction, in: *IEEE Conference on Computer Vision and Pattern Recognition*, 2019, pp. 10512–10521.
- [15] H. Liao, W.A. Lin, S.K. Zhou, J. Luo, ADN: artifact disentanglement network for unsupervised metal artifact reduction, *IEEE Trans. Med. Imaging* (2019), doi:10.1109/TMI.2019.2933425.
- [16] Y. Zhang, H. Yu, Convolutional neural network based metal artifact reduction in X-ray computed tomography, *IEEE Trans. Med. Imaging* 37 (10) (2018) 1370–1381.
- [17] J. Wang, Y. Zhao, J. Noble, B. Dawant, Conditional generative adversarial networks for metal artifact reduction in CT images of the ear, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, 2018, pp. 3–11.
- [18] I. Goodfellow, J. Pouget-Abadie, M. Mirza, X. Bing, D. Warde-Farley, S. Ozair, A. Courville, Y. Bengio, Generative adversarial nets, *Adv. Neural. Inf. Process. Syst.* (2014) 2672–2680.
- [19] M. Arjovsky, S. Chintala, L. Bottou, G.A.N. Wasserstein, in: *International Conference on Machine Learning*, 2017, pp. 214–223.
- [20] Y. Yoshida, T. Miyato, Spectral norm regularization for improving the generalizability of deep learning, *arXiv preprint arXiv:1705.10941*, 2017.
- [21] L. Xu, X. Zeng, Z. Huang, W. Li, H. Zhang, Low-dose chest x-ray image super-resolution using generative adversarial nets with spectral normalization, *Biomed. Signal Process. Control* 55 (2020) 101600.
- [22] C. Ledig, L. Theis, F. Huszar, J. Caballero, W. Shi, Photo-realistic single image super-resolution using a generative adversarial network, in: *IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 105–123.
- [23] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, in: *International Conference on Learning Representations*, 2016, pp. 1–14.
- [24] Y. Ke, X. Wang, L. Le, Z. Ling, R.M. Summers, Deep lesion graphs in the wild: relationship learning and organization of significant radiology image findings in a diverse large-scale lesion database, in: *IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 9261–9270.
- [25] Y. Xie, Y. Xia, J. Zhang, M. Fulham, Y. Zhang, Fusing texture, shape and deep model-learned information at decision level for automated classification of lung nodules on chest CT, *Inform. Fusion* 42 (1) (2018) 102–110.
- [26] A. Levin, D. Lischinski, Y. Weiss, A closed-form solution to natural image matting, *IEEE Trans. Pattern Anal. Mach. Intell.* 30 (2) (2007) 228–242.
- [27] X. Wang, Laplacian operator-based edge detectors, *IEEE Trans. Pattern Anal. Mach. Intell.* 29 (5) (2007) 886–890.
- [28] C. Xian, An edge detection new algorithm based on Laplacian operator, in: *IEEE 3-rd International Conference on Communication software Networks*, 2011, pp. 202–206.
- [29] H. Sheikh, A. Bovik, An information fidelity criterion for image quality assessment using natural scene statistics, *IEEE Trans. Image Process.* 14 (12) (2006) 2117–2128.
- [30] Z. Wang, E. Simoncelli, A. Bovik, Multiscale structural similarity for image quality assessment, in: *IEEE Asilomar Conference on Signals, Systems and Computers*, 2003, pp. 1398–1402.
- [31] Z. Lin, Z. Lei, M. Xuanqin, Z. David, FSIM: a feature similarity index for image quality assessment, *IEEE Trans. Image Process.* 20 (8) (2011) 2378–2386.
- [32] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, G. Klambauer, S. Hochreiter, GANs trained by a two time-scale update rule converge to a Nash equilibrium, in: *Advances in Neural Information Processing Systems*, 2017, pp. 2672–2680.
- [33] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, C. Xi, Improved techniques for training GANs, *Advances in Neural Information Processing Systems*, 2016, pp. 2234–2242.
- [34] T. Karras, T. Aila, S. Laine, J. Lehtinen, Progressive growing of GANs for improved quality, stability, and variation, in: *International Conference on Learning Representations*, 2017, pp. 1–26.
- [35] K. Shmelkov, C. Schmid, K. Alahari, How good is my GAN? in: *IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 3654–3668.
- [36] K. Diederik, B. Jimmy, Adam: a method for stochastic optimization, in: *International Conference on Learning Representations*, 2015, pp. 165–179.
- [37] E. Hoffer, I. Hubara, D. Soudry, Train longer, generalize better: closing the generalization gap in large batch training of neural networks, in: *Advances in Neural Information Processing Systems*, 2017, pp. 1731–1741.
- [38] Z.H. Tang, Y.Y. Li, X.Y. Chai, H.Y. Zhang, S.X. Cao, Adaptive Nonlinear Model Predictive Control of NOx Emissions under Load Constraints in Power Plant Boilers, *J. Chem. Eng. Jpn.* 53 (1) (2020) 36–44 volDOI10.1252/jcej.19we142.
- [39] Bo Hu, Changjiang Zhou, Hongbing Wang, Siyu Chen, Nonlinear tribo-dynamic model and experimental verification of a spur gear drive under loss-of-lubrication condition, *Mech. Syst. Signal. Process.* 153 (2021) 107509.
- [40] S. Reddi, S. Kale, S. Kumar, On the convergence of Adam and beyond, in: *International Conference on Learning Representations*, 2018, pp. 1–23.